

ENEL301 – Fundamentals of Economics and Engineering Management

Bias, Diversity and Ethics



What we are going to cover and when...

Week	Lectures Tuesday 10-11 & Thursday 9-10	Tutorial / Workshop Weeks 1, 4-6 Wednesday 3-4 or 4-5. Weeks 2&3 Wednesday 3-5
1	Introduction to Engineering Practice & Ethics	Ethics
	Collaboration & Teamwork	
2	Māori Perspectives in Engineering	Bicultural Competence and Confidence (timetable and attend one of B01-07 OR B08-14)
	Bias, Diversity & Ethics	
3	Intellectual Property (1/2)	
	Intellectual Property (2/2)	
4	Law in NZ (1/2)	IP Protection
	Law in NZ (2/2)	
5	Data Privacy and Data Ethics	Law, Data Privacy, Data Ethics
	Strategy, Decisions & Risk	
6	Sustainable Development & Systems Thinking	Sustainable Development & Systems Thinking
	Life cycle assessment (1/2)	

The background....

1. Engineers cannot achieve much without a lot of help from other people.
4. We need to do the right thing
 - Social culture shapes us, and therefore our ethics → everyone is different

Your moral compass



"Judge Davenport settled on an end sentence of two years and two months' imprisonment, after allowing 55% in discounts for his guilty plea, previous good character, remorse, rehabilitation efforts and his offer to pay the money back."

<https://www.nzherald.co.nz/nz/crime/auckland-engineer-shyamal-shah-jailed-over-1m-watercare-fraud/6D33OURONNEO5NBD2VWXFZXVIE/>

This graduate likely met this course outcome:

"Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice"



Sounds familiar....?

"Apply ethical principles within engineering contexts, and recognise the social and cultural factors which affect ethics."

Your moral compass

- Ethics might not be teachable (Halbeslebenm 2005) vs
 - Ethical reasoning develops over time and can be taught (Perri et al. 2009)
- Develop awareness of how your decisions can be affected by bias (including ethical decisions)

Your moral compass

What we are going to cover today...

- Cognitive biases
- Diversity of perspectives
- Group dynamics

- Codes of conduct (looking into these in more detail, based on your work last week)

Cognitive biases

- Heuristics
 - Conscious or unconscious cognitive processes that ignore some information; 'rules of thumb' for the brain
 - Simplify decision making
- Cognitive biases
 - A systematic error in reasoning caused by heuristics; a distortion of reality
 - Include stereotypes and implicit (unconscious) bias
 - Affect our interaction with others and decision making
- [Check your bias:](https://implicit.harvard.edu/implicit/takeatest.html)
<https://implicit.harvard.edu/implicit/takeatest.html>



[This Photo](#)

[CC BY-SA-NC](#)

Cognitive biases

Stereotypes...

- A generalised belief about a particular group of people
 - e.g. “Engineers are introverted”
 - The reality.... The ‘average’ engineer has similar levels of extraversion / introversion to the general population (Zacher, 2014, Crossin, 2024)
- Sometimes stereotypes can be true (Ellemers, 2018)



Cognitive biases

Implicit and explicit

- Cognitive bias – a systematic error in reasoning
- Explicit – openly expressed attitudes and beliefs
- Implicit – unconscious or hidden bias
- “On average across 36 countries, there were implicit and explicit attitudes and stereotypes that favo(u)red the same high-status groups (e.g., white, light-skin, thin, young, straight, men) relative to the same low-status groups (e.g., black, dark-skin, fat, old, gay, women)”
(Charlesworth *et al.*, 2023)



anchoring

The first thing you judge influences your judgment of all that follows.

Human minds are associative in nature, so the order in which we receive information helps determine the course of our judgments and perceptions.

Be especially mindful of this bias during financial negotiations such as houses, cars, and salaries. The initial price offered is proven to have a significant effect.



confirmation bias

You look for ways to justify your existing beliefs.

We automatically find ways to make new information fit our existing narratives and preconceptions, and to dismiss information that does not.

Think of your ideas and beliefs as software you actively trying to find problems with rather than things to be defended.

"The first principle is that you must not fool yourself – and you are the easiest person to fool." – Richard Feynman



backfire effect

When your core beliefs are challenged, it can cause you to believe even more strongly.

We can experience being wrong about some ideas as an attack upon our very selves, or our tribal identity. This can lead to motivated reasoning which causes us to reinforce our beliefs even if we might accept particular facts and disconfirming evidence.

"It ain't what you don't know that gets you into trouble. It's what you know for sure that just ain't so."

– Mark Twain



declinism

You see the past as better than it was, and expect the future to be worse than it is likely.

Despite living in the most peaceful and prosperous time in history, many people believe things are getting worse. The 24 hour news cycle, with its reporting of overly negative and violent events, may account for some of this effect.

Instead of relying on nostalgic impressions of how great things used to be, use measurable metrics such as life expectancy, levels of crime and violence, and prosperity statistics.



just world hypothesis

Your preference for a just world makes you presume that it exists.

A world in which people don't always get what they deserve, hard work doesn't always pay off, and injustice happens is an uncomfortable one that threatens our preferred narrative.

However, it is also the reality.

A more just world requires understanding rather than blame. Remember that everyone has their own life story, we're all fallible, and bad things happen to good people.



sunk cost fallacy

You irrationally cling to things that have already cost you something.

When we've invested our time, money, or emotion into something, it hurts us to let it go. This aversion to pain can distort our better judgment and cause us to make unwise investments.

To regain objectivity, ask yourself: had I not already invested something, would I still do so now? What would I counsel a friend to do if they were in the same situation?



dunning-kruger effect

The more you know, the less confident you're likely to be.

Because experts know just how much they don't know, they tend to underestimate their ability, but it's easy to be over-confident when you have only a simple idea of how things are.

"The whole problem with the world is that fools and fanatics are so certain of themselves, yet wiser people so full of doubts."

– Bertrand Russell



barnum effect

You see personal specifics in vague statements by filling in the gaps.

Because our minds are given to making connections, it's easy for us to take nebulous statements and find ways to interpret them so that they seem specific and personal.

Psychics, astrologers and others use this bias to make it seem like they're telling you something relevant. Consider how things might be interpreted to apply to anyone, not just you.



framing effect

You allow yourself to be unduly influenced by context and delivery.

We all like to think that we think independently, but the truth is that all of us are, in fact, influenced by delivery, framing and subtle cues. This is why the ad industry is a thing, despite almost everyone believing they're not affected by advertising messages.

Only when we have the intellectual humility to accept the fact that we can be manipulated, can we hope to limit how much we are. Try to be mindful of how things are being put to you.



in-group bias

You unfairly favor those who belong to your group.

We presume that we're fair and impartial, but the truth is that we automatically favor those who are most like us, or belong to our groups.

Try to imagine yourself in the position of those in our groups; whilst also attempting to be dispassionate when judging those who belong to your in-groups.



fundamental attribution error

You judge others on their character, but yourself on the situation.

If you haven't had a good night's sleep, you know why you're being a bit slow, but if you observe someone else being slow you don't have such knowledge and so might presume them to just be a slow person.

It's not only kind to view others' situations with charity. It's more objective too. Be mindful to also err on the side of taking personal responsibility rather than justifying and blaming.



placebo effect

If you believe you're taking medicine it can sometimes 'work' even if it's fake.

The placebo effect can work for stuff that our mind influences (such as pain) but not so much for things like viruses or broken bones.

Homeopathy, acupuncture, and many other forms of natural 'medicine' have been proven to be no more effective than placebo. Keep a healthy body and bank balance by using evidence-based medicine from a qualified doctor.



halo effect

How much you like someone, influences your other judgments of them.

Our judgments are associative and automatic, and so if we want to be objective we need to consciously control for irrelevant influences. This is especially important in a professional setting.

We're all affected by cultural and personal prejudices.

It's only through becoming aware of them that we can mitigate their effects.



bystander effect

You presume someone else is going to do something in an emergency situation.

When something terrible is happening in a public setting we can experience a kind of shock and mental paralysis that distracts us from a sense of personal responsibility. The problem is that everyone can experience this sense of deindividuation in a crowd.

If there's an emergency situation, presume to be the one who will help or call for help. Be the change you want to see in the world.



availability heuristic

Your judgments are influenced by what springs most easily to mind.

How recent, emotionally powerful, or unusual your memories are can make them seem more relevant. This, in turn, can cause you to apply them too readily.

Try to gain different perspectives and relevant statistical information rather than relying purely on first judgments and emotive influences.



belief bias

If a conclusion supports your existing beliefs, you'll rationalize anything that supports it.

It's difficult for us to set aside our existing beliefs to consider the true merits of an argument. In practice this means that our ideas become impervious to criticism, and are perpetually reinforced.

A useful thing to ask is 'when and how did I get this belief?'

We tend to automatically defend our ideas without ever really questioning them.



groupthink

You let the social dynamics of a group situation override the best outcomes.

Dissent can be uncomfortable and dangerous to one's social standing, and so often the most confident or first voice will determine group decisions.

Rather than openly contradicting others, seek to facilitate objective means of evaluation and critical thinking practices as a group activity.



optimism bias

You overestimate the likelihood of positive outcomes.

There can be benefits to a positive attitude, but it's unwise to allow such an attitude to adversely affect our ability to make rational judgments (they're not mutually exclusive).

If you make rational, realistic judgments you'll have a lot more to feel positive about.



reactance

You'd rather do the opposite of what someone is trying to make you do.

When we feel our liberty is being constrained, our inclination is to resist, however in doing so we can over-compensate.

Be careful not to lose objectivity when someone is being coercive/manipulative, or trying to force you to do something. Wisdom springs from reflection, folly from reaction.



curse of knowledge

Once you understand something you presume it to be obvious to everyone.

Things make sense once they make sense, so it can be hard to remember why they didn't. We build complex networks of understanding and forget how intricate the path to our available knowledge really is.

When teaching someone something new, go slow and explain like they're ten years old (without being patronizing). Repeat key points and facilitate active practice to help embed knowledge.



self-serving bias

You believe your failures are due to external factors, yet you're personally responsible for your successes.

Many of us enjoy unearned privileges, luck and advantages that others do not. It's easy to tell ourselves that we deserve these things, whilst blaming circumstance when things don't go our way.

When judging others, be mindful of how this bias interacts with the just-world hypothesis, fundamental attribution error, and the in-group bias.



negativity bias

You allow negative things to disproportionately influence your thinking.

The pain of loss and hurt are felt more keenly and persistently than the fleeting gratification of pleasant things. We are wired for survival, and our aversion to pain can distort our judgment for a modern world.

Pro-and-con lists, as well as thinking in terms of probabilities, can help you evaluate things more objectively than relying on a cognitive impression.



pessimism bias

You overestimate the likelihood of negative outcomes.

Pessimism is often a defense mechanism against disappointment, or it can be the result of depression and anxiety disorders.

Perhaps the worst aspect of pessimism is that even if something good happens, you'll probably feel pessimistic about it anyway.



spotlight effect

You overestimate how much people notice how you look and act.

Most people are much more concerned about themselves than they are about you. Absent overt prejudices, people generally want to like and get along with you as it gives them validation too.

Instead of worrying about how you're being judged, consider how you make others feel. They'll remember this much more, and you'll make the world a better place.

thou shalt not suffer cognitive biases

Cognitive biases make our judgments irrational. We have evolved to use shortcuts in our thinking, which are often useful, but a cognitive bias means there's a kind of misfiring going on causing us to lose objectivity. This poster has been designed to help you identify some of the most common biases and how to avoid falling victim to them. Help people become aware of their biases generally by sharing the website yourbias.is or more specifically e.g. yourbias.is/confirmation-bias



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Free to download from thethinkingshop.org

Bias in generative AI...

- GPT-4 exhibits racial and gender bias, including in the creation of medical education case studies, diagnostic reasoning, and medical plans (Zack and Lehman, 2024)

Bias in generative AI...

Open AI's terms of service, as at 04/06/2025

<https://openai.com/policies/row-terms-of-use/>

When you use our Services you understand and agree:

- Output may not always be accurate. You should not rely on Output from our Services as a sole source of truth or factual information, or as a substitute for professional advice.
- You must evaluate Output for accuracy and appropriateness for your use case, including using human review as appropriate, before using or sharing Output from the Services.
- You must not use any Output relating to a person for any purpose that could have a legal or material impact on that person, such as making credit, educational, employment, housing, insurance, legal, medical, or other important decisions about them.
- Our Services may provide incomplete, incorrect, or offensive Output that does not represent OpenAI's views. If Output references any third party products or services, it doesn't mean the third party endorses or is affiliated with OpenAI.

Diversity of perspectives

Our characteristics affect our decisions

- *Culture*
- *Experience*
- *Background*
- *Interests*
- *Education*
- *Ethics*
- *Gender*
- *Knowledge*
- *Morals*
- *Neurodivergence*
- *Skills*
- *Upbringing*
- *Physical abilities and disabilities*
- *Extracurricular activities*
- *Maturity*
- *Prior knowledge*
- *Personality*
- etc.*

Teams will always be made up of people with different characteristics.

When we are in teams who are more like us, we are more homogenous. However, diversity will always exist in group decision making

→ Purposefully and positively exploit our differences

Diversity of perspectives

Our characteristics affect our decisions

Biological and physical

- Ability and skills
- Neurodiversity
- Genetic variation (e.g. skin colour, hair colour)
- Sex
- Age

Language and communication

- Spoken and sign
- Non-verbal
- Storytelling traditions
- Communication preferences

Cognitive and emotional

- Intelligence
- Self-awareness
- Creativity
- Knowledge

Social, Cultural Groups

- Culture
- Religion and faith
- Social norms
- Family structures
- Socio-economics
- Education

Personality and identity

- Personality traits
- Experiences
- Gender
- Sexual orientation
- Values and beliefs

Ethical and moral perspectives

- Alignment with ethical theories
- Professional vs personal
- Work vs personal

Diversity of perspectives

Our characteristics affect our decisions

Risk

- Men are generally more likely to take risks than women
 - but other factors affect risk-behaviour (e.g. age, type of risk; Byrnes, Miller & Schafer [2019])

Creativity

- Age was not significantly related to creativity. Age was neither related to employee creativity as rated by supervisor nor to self reports of creativity (The relationship of age to ten dimensions of job performance, Ng & Feldman, 2008)

Diversity of perspectives

Our characteristics affect our decisions

Culture is “... a system of shared beliefs, values, behavio(u)rs and practices common to a particular group or population that results from group experience..” – Di Mattia, 2008

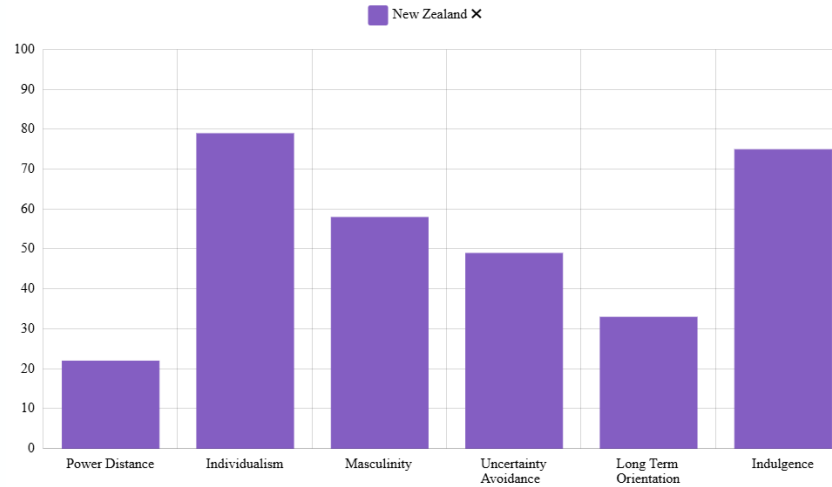
- Hofstede’s model of culture
 - Power Distance
 - Individualism/Collectivism
 - Uncertainty Avoidance
 - Masculinity/Feminity

Later additions:

- Long-term orientation
- Indulgence
- Note: Hofstede’s model is disputed – your homework!

Diversity of perspectives

Our characteristics affect our decisions



<https://geerthofstede.com/country-comparison-bar-charts/>

- How might low power distance and high individualism affect ethical decision making?
- Living in countries with high collectivism is associated with less age bias (Ackerman and Chopik, 2021)
- Group bias increases in more uncertainty avoiding contexts (Fisher and Derham, 2016)

Diversity of perspectives

Cultural perspectives; Ngāi Tahu values

Whanaungatanga (family); We will respect, foster and maintain important relationships within the organisation, within the iwi and within the community.

Manaakitanga (looking after our people); We will pay respect to each other, to iwi members and to all others in accordance with our tikanga (customs).

Tohungatanga (expertise); We will pursue knowledge and ideas that will strengthen and grow Ngāi Tahu and our community.

Kaitiakitanga (stewardship); We will work actively to protect the people, environment, knowledge, culture, language and resources important to Ngāi Tahu for future generations.

Tikanga (appropriate action); We will strive to ensure that the tikanga of Ngāi Tahu is actioned and acknowledged in all of our outcomes.

Rangatiratanga (leadership); We will strive to maintain a high degree of personal integrity and ethical behaviour in all actions and decisions we undertake.

<https://ngaitahu.iwi.nz/te-runanga-o-ngai-tahu/>

Group dynamics

“...changes that may occur in any part of the group and bring out actions and reactions in the group structure that affects group members.”

Factors that affect group dynamics include:

- Leadership
- Group norms: “Rules” of behaviour & social pressure and conformity
- Peoples attitudes and behaviour
 - is different in a group AND
 - are affected by the group

Group dynamics & conformity

The Asch Experiments



Group dynamics & conformity



Group dynamics & conformity

Groupthink symptoms

“when concurrence-seeking ... override(s) realistic appraisal of alternative courses of action.” – I.L. Janis (2008)

1. Illusion of invulnerability
2. Collective rationalisation
3. Belief in inherent morality
4. Stereotyped views of external groups
5. Direct pressure on dissenters
6. Self-censorship
7. Illusion of unanimity
8. Self-appointed ‘mindguards’

(I.L. Janis, 1982)

Group dynamics & conformity

Groupthink outcomes



Crew of the final ill-fated flight of the Space Shuttle "Columbia", mission STS-107. Source:

<http://www.af.mil/shared/media/photodb/photos/030201-F-9999G-001.jpg> (open source)

“...program managers **created huge barriers against dissenting opinions** by stating preconceived conclusions based on subjective knowledge and experience, rather than on solid data.”

Harold Gehman (Chair), *Columbia Accident Investigation Report*, August 2003.

Group dynamics & conformity

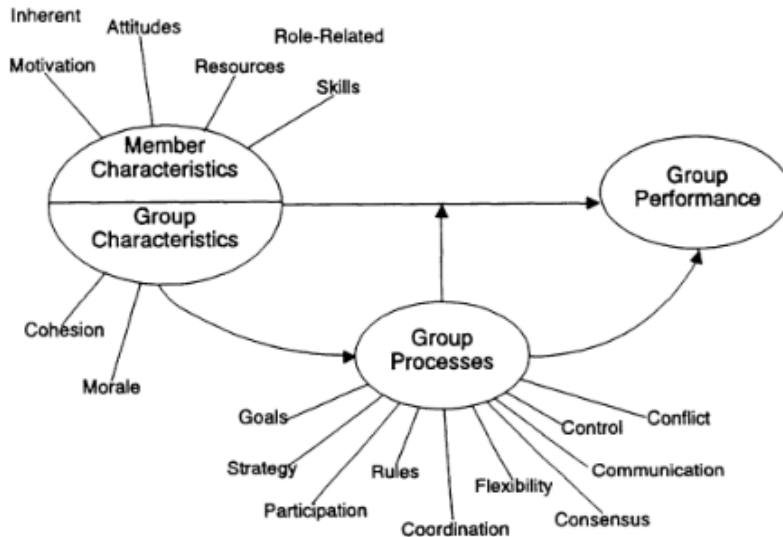
- Cohesion can lead to better decision making and performance. However...
- High group cohesiveness is one of the conditions for groupthink
- Individual dissent can lead to:
 - Dissenter's opinions moving towards those of the group OR
 - Rejection of the dissenter (and dissenting opinion) by the group

Group leader:

- facilitate communication, including dissenting voices
- create an environment for
 - opinions to be aired
 - mutual respect for different views

Group dynamics & conformity

Decision making and performance



(Maznevski, 1994)

- More diverse teams can perform better
- However... there is also evidence to the contrary - diverse teams don't always perform well!
- One key factors for team performance : collaboration and interaction processes, enabled by excellent communication (Maznevski, 1994)
- Group decisions are ultimately made by individuals

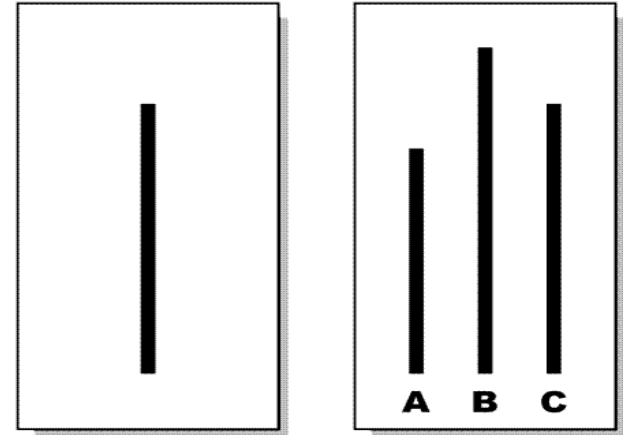
Group dynamics & conformity

Ethical decisions

How might group conformity affect ethical decision making?

How can we address conformity?

- Framework for decision making (e.g. Codes of Conduct / Ethics)
- Normalise dissent (Boeing used to do this...)
- Anonymity / anonymous input
- Diverse perspectives, including externals

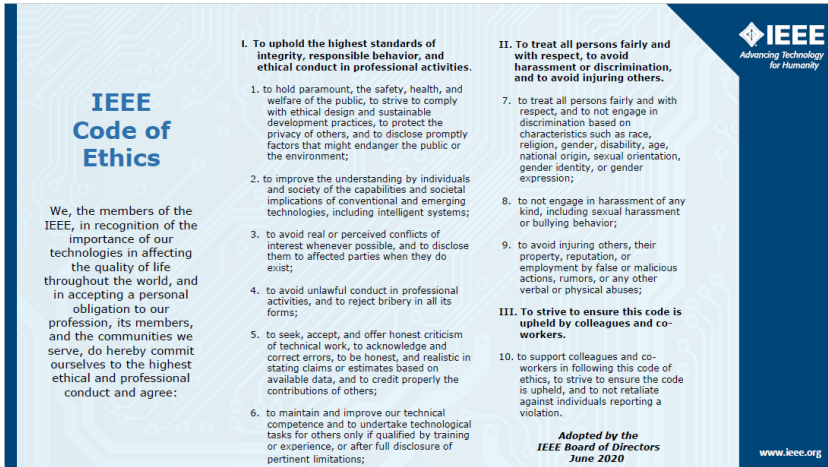


Individually

- Call it out. If you are challenging it internally, chances are someone else is too.
- Consequences of questioning the majority vs keeping quiet?

Professional societies & codes of conduct

Ethical decisions



- Codes of conduct:
 - Written by a group of people, typically with strong input from members and experts
 - Set clear and collective standards and expectations
 - (Should have) accounted for diverse perspectives
- How should this one be improved (any insights from Week 1 Workshop?)

Take-away messages

Cognitive Biases:

- Stereotypes and implicit biases affect decision-making

Diversity:

- Diverse perspectives support better decisions

Group Dynamics:

- Conformity, groupthink, individual and leader behaviour affect decisions

Codes of Conduct:

- Set the ethical standard, based on diverse input

Homework to reinforce learning

- Complete the Questionnaire on LEARN
- Download and read:
 - Jones, M. (2007). Hofstede-culturally questionable?
<https://ro.uow.edu.au/ndownloader/files/50565060>
 - Answer the following questions:
 - Should Hofstede's model be applied to organisations? Why?
 - Should Hofstede's model be applied to individuals? Why?

Complete the Questionnaire on LEARN

By 9 am August 10....

Under the **Bias, Diversity and Ethics** Section

Sensors installed on smartphones, smartwatches, and other wearable devices can collect data that can be used to assess health conditions. You have been gifted a high-end smartwatch from Zealiq, a New Zealand-based private company.

The Zealiq smartwatch includes sensors to detect pulse, heart-rate, bloody oxygen levels, location data, and physical movement. The smartwatch also has capability to run and share data with popular apps, including Spotify and Snapchat, and with fitness apps such as MyFitnessPal and Strava.

Zealiq's end-user agreement indicates that Zealiq will use your data to provide you with personalised health recommendations, including insights related to the early detection of high-risk conditions (e.g. cancer). Zealiq guarantees that your data will be safe, anonymous, and protected from misuse. With your consent, your personal data from the Zealiq watch will be transmitted Zealiq for analysis.

Would you accept this end user agreement?

- ☐ Yes, I would accept the end user agreement and share my data with Zealiq.
- ☐ Maybe, it depends on the detail in the end user agreement.
- ☐ No, I would not accept the end user agreement.

What's next?

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6	Sustainable Development & Systems Thinking	Sustainable Development & Systems Thinking
	Life cycle assessment (1/2)	

Your teaching team

Bicultural Confidence and Competence

Week	Session	Room	Tutor
Week 3 Wed 29 July 15:00 - 17:00	B-08	Rātā E12	Stella Keenan
	B-09	Jack Erskine 340	Hēmi McNeill
	B-10	Jack Erskine 101	Sam Draper
	B-11	Jack Erskine 111	Lydia Frater
	B-12	Jack Erskine 446	Gabby Rosemergy
	B-13	Meremere 526	Kahu Jones
	B-14	Rātā 222 & 223	Benjamin Cosford

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